(PV-SV-02) Software Development Planning

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Document Approval

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Revision History

Rev.	Date	Description
`0.0`	`2018-10-08`	Firstly prepared
`0.1`	`2023-05-10`	Updated according to IEC 62304:2015

Rev.	Date	Description
`0.2`	`2023-12-18`	Document number changed from 603 to Z01 according to OP-709
`0.3`	`2025-08-14`	Added software maintenance and problem-resolution process in section 10

1. Purpose

This document describes the overall software development process used for Portview and defines the planning controls applied to development, verification, configuration management, and maintenance activities.

The plan is intended to:

- define the software lifecycle and its managed phases
- identify development roles, responsibilities, and controlled deliverables
- define verification, risk-management, and maintenance planning expectations
- define configuration-control and naming conventions for the Portview document set

2. Scope

This plan applies to the Portview software lifecycle, from development planning and implementation through verification, release, maintenance, and problem resolution.

In scope:

- lifecycle planning for Portview software activities
- controlled deliverables and document naming conventions
- software verification and validation planning
- software risk-management planning
- software configuration management and change control
- maintenance and post-release problem-resolution process

3. Referenced Documents

The following references govern this planning document.

Reference	Use
Medical Device Regulation (EU) 2017/745	Regulatory framework
Quality management systems - Requirements	Quality-system reference
EN ISO 13485:2016 Medical devices Quality management systems	Governing quality management standard
EN 60601-1:2006/A1:2013 Medical electrical equipment	Basic safety and essential performance reference
ISO 12052:2006 DICOM including workflow and data management	Governing interoperability standard
IEC 62304:2006/AMD1:2015 Software lifecycle processes	Governing software lifecycle standard
Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices	Regulatory guidance
General Principles of Software Validation Final Guidance for Industry and FDA Staff	Regulatory guidance
Software and Medical Devices (VdTUV, 2001)	External guidance reference
QM Quality System Documentation	Internal quality reference
QM Design Control Procedure	Internal process reference
QM Software Development Procedure	Internal process reference
QM Corrective and Preventive Actions	Internal process reference
QM Document Control	Internal process reference

4. Roles And Responsibilities

The software development process is managed through defined organizational roles.

| Activity Area | Responsible Position | Primary Responsibility | | --- | --- | | Project management | CTO (Director) of R&D Center; Deputy General Manager of R&D Center | Approves lifecycle direction, deliverables, and release readiness | | Hazard analysis | Deputy General Manager of R&D Center | Maintains risk-analysis coordination and risk-control review | | Verification and validation testing | Staff of

R&D Center; Manager of R&D Center; Deputy General Manager of R&D Center | Plans, executes, reviews, and closes software verification activities | | Documentation review | Deputy General Manager of R&D Center | Reviews controlled technical documents and revision updates | | Software team | Manager and engineers of R&D Center | Implements software changes, maintains source control, and prepares technical deliverables |

5. Software Lifecycle Overview

Portview follows a managed lifecycle comprising definition, design, implementation, qualification, release, and maintenance activities.

Lifecycle Phase	Planning Intent
Definition	Establish scope, intended use, requirements, and risk inputs
Design	Define software architecture, interfaces, and controlled design outputs
Implementation	Build and maintain the controlled software configuration
Qualification	Perform unit, integration, and system verification activities
Release	Confirm completion of required lifecycle outputs and approved release content
Maintenance	Control released changes, bug fixes, upgrades, and post-release feedback

5.1 Lifecycle Governance

The lifecycle is governed through controlled inputs, documented outputs, and review gates at the end of each major phase.

Governance Topic	Planning Statement
Entry to a phase	Approved inputs, assigned owners, and available controlled templates are required
Exit from a phase	Required deliverables, review findings, and open actions are recorded before closure

Governance Topic	Planning Statement
Change during a phase	Changes are managed through the approved configuration-management and change-control process
Release readiness	Release requires completed verification evidence, dispositioned anomalies, and approved records

6. Development Processes And Deliverables

The Portview software process uses controlled phases, end-of-phase reviews, technical documentation, and lifecycle deliverables.

Topic	Planning Statement
Lifecycle process	Activities and tasks are conducted under a controlled software lifecycle
End-of-phase review	Each major phase is expected to close with review of deliverables and open actions
Technical documentation	Controlled documents are maintained for requirements, design, verification, traceability, and release evidence
Traceability	Relationships between system requirements, software requirements, tests, and risk controls are maintained through the document set

6.1 Controlled Portview Document Set

Document Type	Controlled Identifier Pattern
Software Validation Report	`PV-SV-01`
Software Development Planning	`PV-SV-02`
Software High Level Design	`PV-SV-03`
Software Verification Plan	`PV-SV-04`

Document Type	Controlled Identifier Pattern
Software Verification Report	`PV-SV-05`
Traceability Matrix	`PV-TM-01`
Requirement Specification	`PV-RS-01`
Software Requirement Specification	`PV-SRS-01`
Software Design Specification	`PV-SDS-01`
Software Unit Test Procedure / Result	`PV-STP-01`, `PV-STR-01`
Software Integration Test Procedure / Result	`PV-TP-01`, `PV-TR-01`
System Test Procedure / Result	`PV-SYSTP-01`, `PV-SYSTR-01`

6.2 Planned Deliverables By Lifecycle Stage

Lifecycle Stage	Primary Deliverables
Definition	Requirement inputs, risk inputs, planning records
Design	High-level design, software design specifications, architecture traceability
Implementation	Controlled source code, build outputs, code-review records
Qualification	Unit, integration, and system procedures and results
Release	Validation report, release record, approved anomaly disposition
Maintenance	Change requests, change instructions, maintenance verification records

7. Verification And Validation Planning

Verification and validation activities are planned by level and tied to controlled procedure sets.

Verification Level	Planning Statement
Unit test	Software units are verified against detailed requirement inputs and acceptance criteria
Integration test	Integrated software behavior is evaluated across software-unit interfaces
System test	Software system behavior is evaluated against system-level requirements before release

7.1 Verification Planning Principles

The Portview verification plan is expected to ensure that:

- each verification level is tied to controlled inputs and expected outputs
- acceptance criteria are established before execution
- regression testing is applied when software changes affect released behavior
- verification status is recorded through controlled execution and result records

7.2 Acceptance Criteria

The acceptance criteria for Portview software verification are intended to ensure that requirements:

- implement system-level requirements including those related to risk mitigations
- do not contradict one another
- are expressed in terms that avoid ambiguity
- permit establishment of test acceptance criteria and objective evaluation
- are uniquely identified
- are traceable to design outputs and verification evidence

7.3 Development Tool Validation

Development-support tools used in the Portview software lifecycle are validated or qualified as follows:

Tool	Version	Purpose	Validation Approach
Microsoft Visual C++	2017 Professional Edition	C++ compilation and build	Generally recognized OTS development tool; validated through industry adoption and build-output verification
Git server	Controlled configuration	Source control and change management	Validated through established use in the controlled development environment
Genoray testing platform	Default configuration	Test execution environment	Validated through controlled hardware configuration and documented test setup

Tool validation records are maintained in the QMS controlled archive. If a development-support tool is upgraded or replaced, the tool-validation assessment is repeated before the tool is used in controlled verification activities.

8. Risk Management Planning

Software safety requirements are flowed down from the system risk-management files. Portview units are assessed through risk analysis and FMEA, and risk-control measures are verified through the controlled verification and validation activities.

Risk Topic	Planning Statement
Risk input	Flowed down from system risk-management files
Analysis method	Function/module-based risk review and FMEA
Mitigation verification	Verified through controlled V&V activities
Residual risk handling	Evaluated through the controlled risk-management process

9. Documentation And Naming Conventions

Every controlled document should maintain a defined identifier and revision convention.

Rule	Description
Prefix	Controlled document type identifier
Suffix	`Rev <n>.<n>` or the applicable controlled revision format
Naming pattern	Portview document names should follow the controlled product and document-code convention

9.1 Documentation Control Expectations

Topic	Expectation
Revision control	Each controlled document retains an approved revision history
Naming consistency	Product naming, document identifiers, and revision references remain consistent across the document set
Distribution	Controlled documents are released through the approved document-control process
Traceability references	Cross-references should use controlled identifiers rather than informal names where possible

10. Configuration Management And Change Control

Portview source and related software subsystems are controlled through a software configuration-management system and change-control process.

Configuration Topic	Planning Statement
Source control	Portview source is controlled through a Git server with change control
Configuration items	Software documents, builds, procedures, results, and supporting artifacts are placed under control

Configuration Topic	Planning Statement
Change management	Changes are processed through the approved design and software change procedures
Build integrity	Engineering management is responsible for ensuring implementation integrity within the controlled configuration baseline

10.1 Controlled Configuration Items

Configuration Item	Purpose
Validation report	Validation effectiveness evidence
Development planning	Lifecycle planning and task definition
High-level design	Architecture and item decomposition
Verification plan	Unit, integration, and system test planning
Verification report	Released verification outcomes
Requirement and design specifications	Controlled technical inputs and outputs
Test procedures and results	Released verification records

10.2 Configuration-Control Expectations

Topic	Expectation
Baseline identification	Each released build and related document set is associated with a controlled baseline
Change approval	Software changes require review and approval before release
Status accounting	Configuration status is maintained for source, documents, procedures, and released outputs
Archiving	Released software and supporting records are retained in the approved archive structure

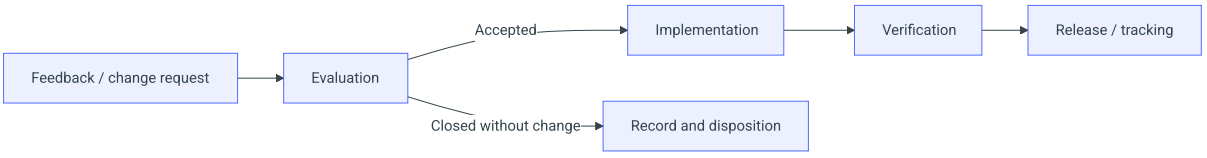
11. Software Maintenance And Problem Resolution

Post-release feedback and software change handling are managed through a controlled maintenance and problem-resolution process.

Topic	Process Summary
Feedback intake	Service and sales teams receive requests and prepare change records
Evaluation	R&D/Product Management evaluates whether a request is a problem to resolve
Resolution	Maintainers implement approved changes and report completion through change instructions
Tracking	Requests and resulting instructions are tracked under change control
Post-release risk management	Risk-management activities are applied to released-software changes when applicable
SOUP maintenance	Upgrades, bug fixes, patches, and obsolescence handling are managed through the controlled process

11.1 Maintenance Decision Flow

The post-release process is expected to distinguish between feedback that requires correction, feedback that results in an approved change request, and feedback that is closed without software modification.



11.2 Security Patch And SOUP Obsolescence Management

Topic	Criteria
Patch classification	Security patches are classified as Critical (immediate deployment), High (deployment within 30 days), or Normal (next scheduled release)

Topic	Criteria
Critical patch trigger	Confirmed exploitable vulnerability in Portview or its SOUP components affecting patient data integrity, system availability, or regulatory compliance
Deployment process	Critical and High patches follow the standard change-control process with expedited verification; Normal patches are bundled into scheduled releases
SOUP obsolescence handling	SOUP components are monitored for end-of-life and security advisories; obsolete components are replaced or isolated before the next scheduled release
Notification	Affected vendors and users are notified of Critical patches via the help desk team; patch deployment instructions are provided with each notification
Verification	Security patches are verified through the applicable unit, integration, or system verification procedures before deployment; regression testing is applied when the patch affects released behavior